Evaluation of $I = \int_0^1 \cos(x^2 + e^x)\,dx$

1 Introduction

We are looking to evaluate

$$I = \int_0^1 \cos(x^2 + e^x)\,dx.$$

No elementary antiderivative exists, but we can obtain an exact expression in the form of a double series.

2 Power series expansion

For any real number u,

$$\cos u = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}\,u^{2n}.$$

By setting $u = x^2 + e^x$, we have

$$\cos(x^2 + e^x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}\,(x^2 + e^x)^{2n}.$$

3 Binomial expansion

For each n, let us expand $(x^2 + e^x)^{2n}$ using the binomial theorem:

$$(x^2 + e^x)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} (x^2)^k (e^x)^{2n-k} = \sum_{k=0}^{2n} \binom{2n}{k} x^{2k} e^{(2n-k)x}.$$

4 Series/integral interchange

The series converges normally on $[0, 1]$, so we can integrate term by term:

$$I = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \sum_{k=0}^{2n} \binom{2n}{k} \underbrace{\int_0^1 x^{2k} e^{(2n-k)x}\,dx}_{I_{k,n}}.$$

5 Exact calculation of $I_{k,n}$

Let $m = 2k$ and $a = 2n - k$. We must evaluate

$$I(m, a) = \int_0^1 x^m e^{ax} \, dx.$$

- If $a = 0$ (i.e., $k = 2n$): $I(m, 0) = \dfrac{1}{m+1} = \dfrac{1}{4n+1}$.
- If $a \neq 0$, we integrate by parts m times or use the explicit formula

$$\begin{aligned} I(m, a) &= \frac{m!}{(-a)^{m+1}} \left(1 - e^a \sum_{j=0}^{m} \frac{(-a)^j}{j!} \right) \\ &= \frac{(2k)!}{(k-2n)^{2k+1}} \left(1 - e^{2n-k} \sum_{j=0}^{2k} \frac{(k-2n)^j}{j!} \right). \end{aligned}$$

6 Exact expression of the integral

By substituting both cases into the double sum, we obtain the exact representation:

$$\begin{aligned} I = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \Bigg[& \binom{2n}{2n} \frac{1}{4n+1} \\ & + \sum_{\substack{k=0 \\ k \neq 2n}}^{2n} \binom{2n}{k} \frac{(2k)!}{(k-2n)^{2k+1}} \left(1 - e^{2n-k} \sum_{j=0}^{2k} \frac{(k-2n)^j}{j!} \right) \Bigg]. \end{aligned} \tag{1}$$

This expression is exact and only involves factorials, binomial coefficients, and the exponential e.

7 Numerical evaluation

7.1 Direct evaluation

Using an adaptive quadrature method (Gauss-Kronrod), we find

$$I \approx -0.303666963947415 \quad (\text{estimated error } \sim 10^{-15}).$$

7.2 Evaluation of the exact series

The naive calculation of (**??**) in double precision fails quickly: the terms e^{2n-k} and the exponential sums are very large, and their subtraction causes catastrophic cancellation. For $n \geq 10$, the numerical results blow up.

Using multi-precision arithmetic (50 decimal places) and a robust numerical integration of each $I_{k,n}$ (rather than the unstable closed formula), the double series converges to:

$$I = -0.30366696394741517125401403601 5\ldots$$

The two approaches match up to machine precision, validating the exact formula.

8 Retained approximate value

$$\boxed{\int_0^1 \cos(x^2 + e^x)\,dx \simeq -0.303666963947415}$$

with an uncertainty of less than 10^{-14}.

Remark

The expression (**??**) provides an **exact value** of the integral, although it cannot be reduced to an elementary constant. For practical calculations, direct numerical integration is preferred.